NETWORK SECURITY

Week 3 - Lecture 3

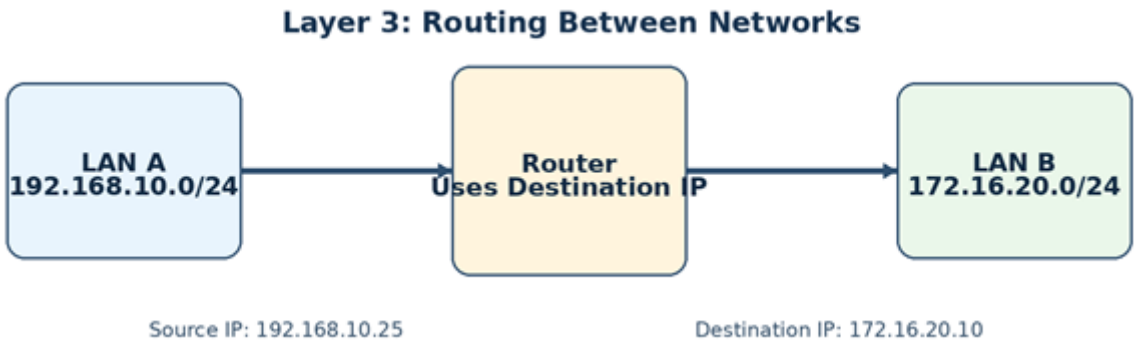
Layer 3: IPv4 and IPsec Security and Protection

Department: Cybersecurity Engineering Techniques | Stage: Fourth Year

Main idea: Layer 3 moves packets between networks. IPv4 delivers packets, while additional controls such as ACLs, filtering, and IPsec provide security.

Learning Outcomes

- Explain the role of Layer 3, IPv4 addressing, and router forwarding.
- Recognize important IPv4 header fields used in delivery and security.
- Describe IP spoofing, ICMP abuse, DoS/DDoS, fragmentation, and routing attacks.
- Explain ACLs, ingress/egress filtering, uRPF, and segmentation.
- Compare AH and ESP and explain IPsec transport and tunnel modes.
- Describe Security Associations (SAs) and the basic role of IKEv2.
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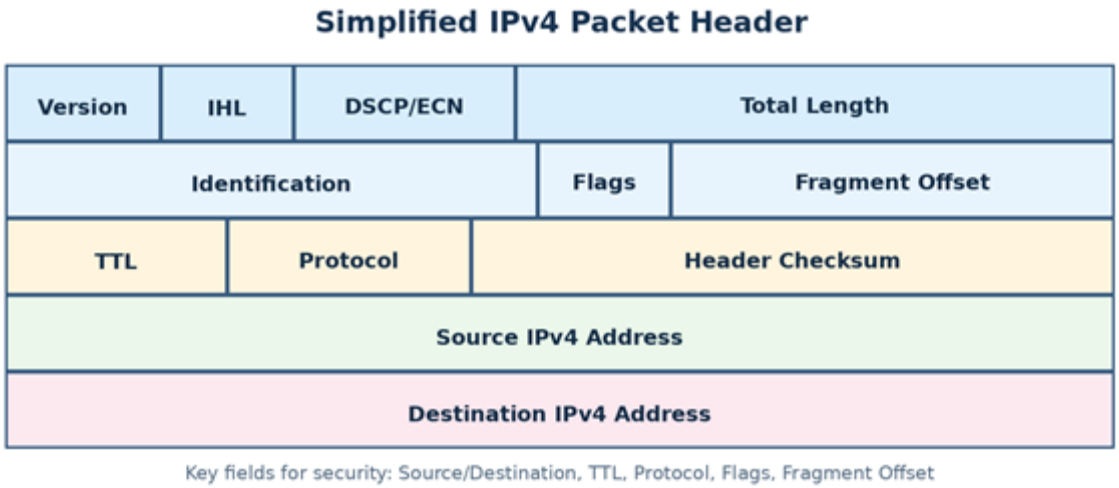
Layer 2 mainly handles local delivery; Layer 3 allows packets to travel between different networks.

1. IPv4 Basics and Packet Structure

IPv4 is a connectionless Network Layer protocol. An IPv4 address contains 32 bits and is normally written in dotted-decimal form, for example 192.168.10.25.

What does a router use?

- Routers primarily examine the destination IPv4 address and forward the packet toward the destination network.
- The source address identifies the device that claims to have sent the packet.



This diagram represents the internal structure of an "IPv4 Packet Header," divided into boxes, where each box corresponds to a field containing specific data transmitted across the network.

The total width of this diagram is 32 bits, and each row represents a portion of the digital packet's header details. Below is a breakdown of the fields shown in the image:

□ Details of the first row (blue boxes)

• Version: Specifies the version of the Internet Protocol being used; here, the value is always set to (4) because we are using IPv4.

• IHL (Internet Header Length): Indicates the size of the header itself (marking where the header ends and the actual data begins), as the header size can sometimes vary if additional protocol features are included.

• DSCP/ECN (Quality of Service): Fields used to determine the priority of a packet (for example, giving higher priority to video or voice call packets compared to web-browsing traffic to ensure a smooth, uninterrupted connection).

• Total Length: Specifies the size of the entire packet (the size of the header plus the data contained within it).

Breakdown of the second row fields

• Identification, Flags, and Fragment Offset: These three fields work together as a set. If a packet is too large for the network, it is broken into smaller pieces (fragmented). These fields act as "sequence numbers and markers," allowing the receiving device to reassemble the fragments in the correct order.

♡ Breakdown of the colored fields at the bottom (green, pink, and yellow)

• TTL (Time to Live): A protective counter that prevents a packet from circulating endlessly across the internet.

It is a digital counter placed within a packet’s header to prevent it from being lost in the network indefinitely.

- The packet begins with a specific number (e.g., 64 or 128).

- Each time the packet passes through a router, the router decrements this number by 1.
- If the packet encounters an issue and ends up circulating in an infinite loop until the counter reaches 0, the packet is immediately deleted and discarded to prevent it from causing congestion on the internet.

• Protocol: Specifies the program or protocol that will receive the packet at the next layer (such as TCP or UDP).

• Header Checksum: A mathematical code used to verify the integrity of the header and ensure it has not been corrupted during transmission.

• Source IPv4 Address: The address of the device that sent the packet (consists of 32 bits).

• Destination IPv4 Address: The address of the target device to which the packet is being sent (consists of 32 bits).

Field	Security / delivery importance
Source & Destination	Identify claimed sender and intended receiver.
TTL	Decreases at each router; prevents packets circulating forever.
Protocol	Identifies upper-layer content such as Internet Control Message Protocol (ICMP), Transmission Control Protocol (TCP), User Datagram Protocol (UDP), Encapsulating Security Payload (ESP), or Authentication Header (AH).
Flags + Fragment Offset	Control fragmentation and reassembly.
Header Checksum	Detects errors in the IPv4 header.
Remember: the router forwards mainly by destination address; a source IP address alone is not strong proof of identity.	

2. Why IPv4 Requires Security

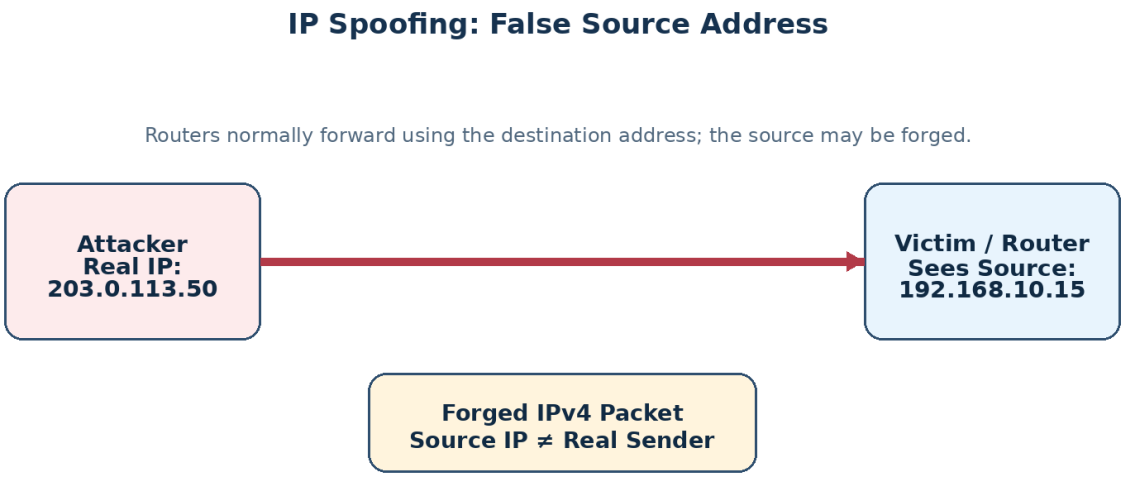
Ordinary IPv4 was designed primarily for packet delivery. It does not automatically encrypt packet contents or strongly authenticate the source address.

Threat	What can happen?
IP spoofing	A false source IPv4 address is placed in a packet.
Packet interception	Unencrypted information may be read on an untrusted path.
ICMP abuse	Ping sweeps, reconnaissance, excessive traffic, or false control messages.
DoS / DDoS	Bandwidth, router capacity, or server resources may be exhausted.
Fragmentation attacks	Unusual fragments may confuse filters or consume reassembly resources.
Routing attacks	Traffic may be redirected, dropped, intercepted, or

	sent through a wrong path.

IP Spoofing

IP spoofing occurs when an attacker creates a packet containing a false source IPv4 address. It may be used to hide the real sender, imitate another device, bypass simple IP-based restrictions, or support DoS attacks.

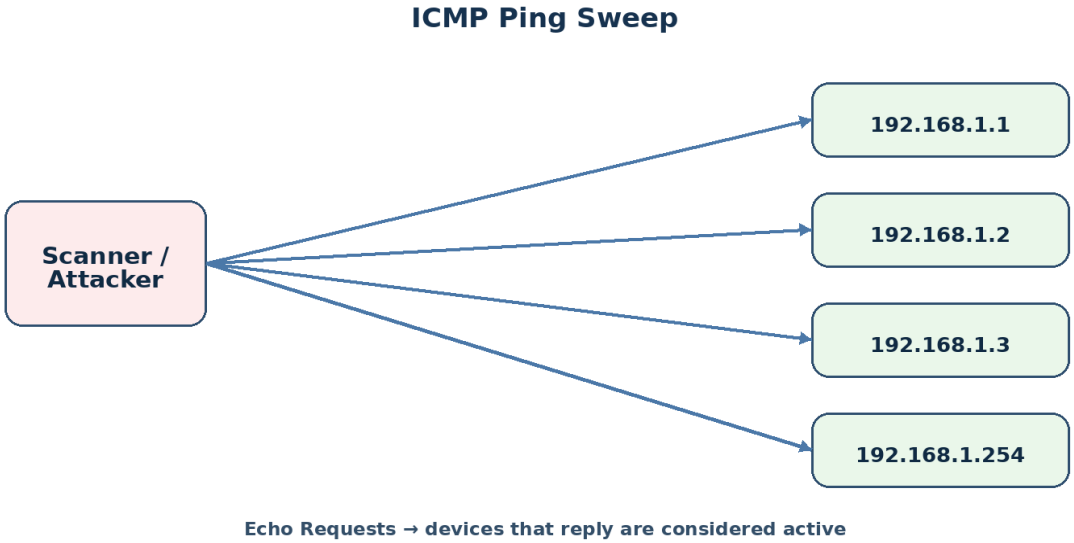


A source IPv4 address should not be treated as strong authentication.

3. ICMP, DoS/DDoS, and Fragmentation Attacks

ICMP: Useful but Abusable

ICMP supports diagnostics and error reporting. Ping uses ICMP Echo Request/Echo Reply, while traceroute also depends on IP and ICMP responses.



Topic	Short explanation	Protection idea
Ping sweep	ICMP requests are sent to many addresses to discover active hosts.	Permit required ICMP, limit excessive traffic, log

		suspicious activity.
DoS / DDoS	Large volumes of packets or requests make a service unavailable.	Filtering, rate limiting, monitoring, and upstream protection.
Fragmentation attack	Very small, overlapping, or incomplete fragments may confuse devices.	Drop malformed fragments, inspect/reassemble fragments, update devices.
Do not blindly block all ICMP: some ICMP messages are important for diagnostics and correct network operation.		

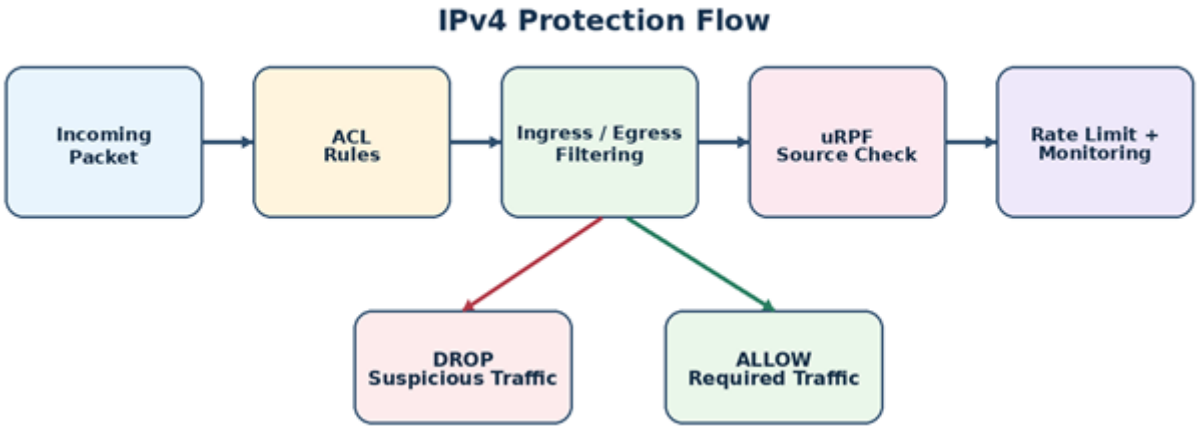
4. Routing Attacks and IPv4 Protection

Routing Attacks

Routers depend on routing information. If routing information or router configuration is changed, traffic may be redirected, dropped, intercepted, or sent through an incorrect path.

- Possible attacker actions: advertise a false route, obtain administrator credentials, send unauthorized routing messages, or create routing loops.
- Protect routers with strong administrator authentication, Secure Shell (SSH), restricted management addresses, routing-protocol authentication, updates, backups, logging, and disabled unnecessary services.

Main IPv4 Protection Mechanisms



Control	Purpose
ACLs	Permit or deny traffic according to policy.
Ingress / Egress filtering	Reduce packets using false or inappropriate source addresses.
Unicast Reverse Path Forwarding (uRPF)	Checks whether the source address is reachable through an expected path.

Segmentation	Separates user groups and sensitive resources.
Rate limiting + monitoring	Reduces abuse and provides visibility.
IPsec	Adds encryption, integrity, authentication, and anti-replay protection.

5. ACLs, Filtering, uRPF, and Segmentation

Access-Control Lists (ACLs)

An ACL is an ordered set of permit or deny rules. Depending on the ACL type, rules can examine source address, destination address, protocol, port number, or ICMP type.

Standard ACL	Extended ACL
Mainly checks the source IPv4 address.	Can check source, destination, protocol, and TCP/UDP port.
Example: permit 192.168.10.0/24	Example: permit HTTPS from 192.168.20.0/24 to 172.16.10.10

ACL Processing Principles

- Rules are checked in order.
- The first matching rule is applied.
- Specific rules should normally appear before general rules.
- Permit only required traffic and test rules before deployment.

Anti-Spoofing Controls

Control	Simple meaning
Ingress filtering	Check traffic entering the network; reject impossible or suspicious source addresses.
Egress filtering	Check traffic leaving the network; prevent internal systems from using unauthorized source addresses.
uRPF (**Unicast Reverse Path Forwarding**)	Compare the source address with the router's expected return path before accepting the packet.

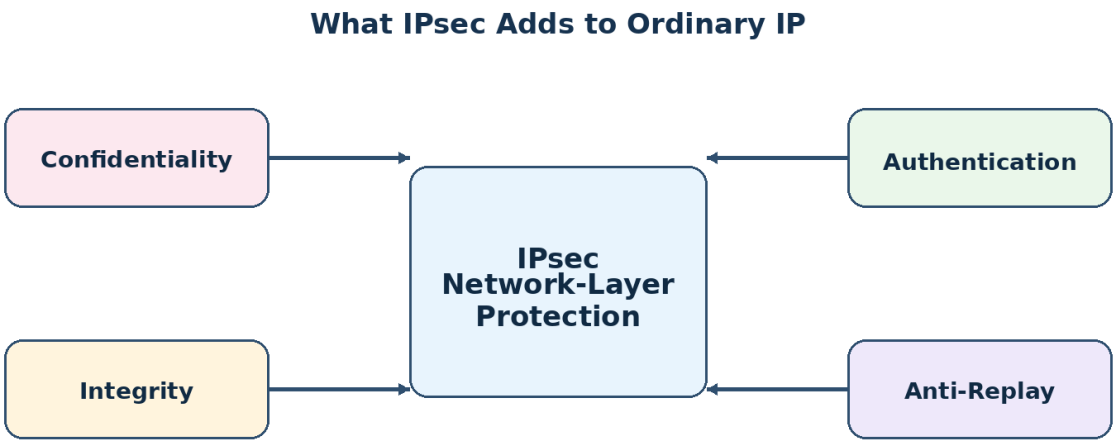
Network Segmentation

Separate students, staff, administration, servers, and management networks.
Segmentation limits unnecessary communication and reduces the effect of attacks.

No single Layer 3 control stops every attack. Use several controls together.

6. Introduction to IPsec

IPsec protects IP communication at the Network Layer. It is useful when packets must cross untrusted networks such as the public Internet.



Access control is applied according to the organization's security policy.

Main Components

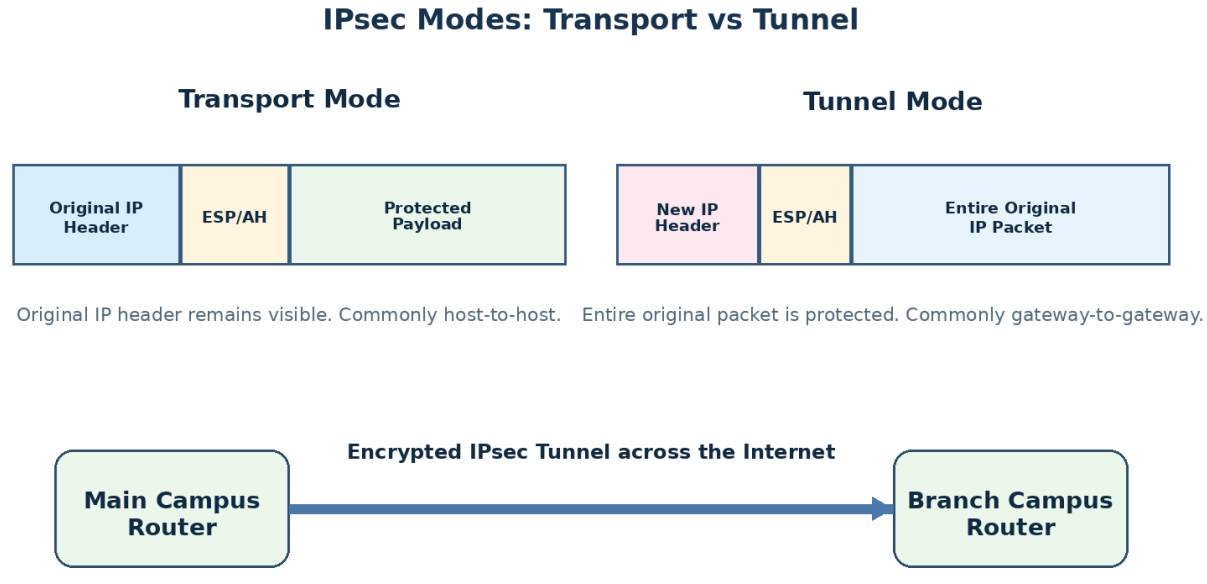
Component	Main role
AH - Authentication Header	Integrity, data-origin authentication, and anti-replay protection. No confidentiality.
ESP - Encapsulating Security Payload	Can provide confidentiality through encryption, plus integrity, authentication, and anti-replay.
Security Association (SA)	Describes how traffic is protected: algorithms, keys, peer, lifetime, SPI, and anti-replay information.
IKEv2	Authenticates peers, negotiates security parameters, establishes keys, and creates/manages SAs.

AH vs ESP

Feature	AH	ESP
Confidentiality	No	Yes
Integrity	Yes	Yes, when configured
Source authentication	Yes	Yes, when configured
Anti-replay	Yes	Yes
Common protected VPN traffic	Limited	Commonly used

7. IPsec Transport Mode and Tunnel Mode

IPsec can protect traffic in two main modes. The difference is how much of the original IP packet is protected and whether a new outer IP header is added.



Feature	Transport Mode	Tunnel Mode
Protection	Mainly protects the original payload.	Protects the entire original IP packet.
IP header	Original IP header remains visible.	A new outer IP header is added.
Typical use	Host-to-host / direct endpoint protection.	Gateway-to-gateway, site-to-site VPN, remote access.
Overhead	Lower	Higher

For a university link between two buildings across the Internet, tunnel mode is the typical choice.

Standard visual references for these concepts are also available from Wikimedia Commons (IPv4 Header and IPsec Modes).

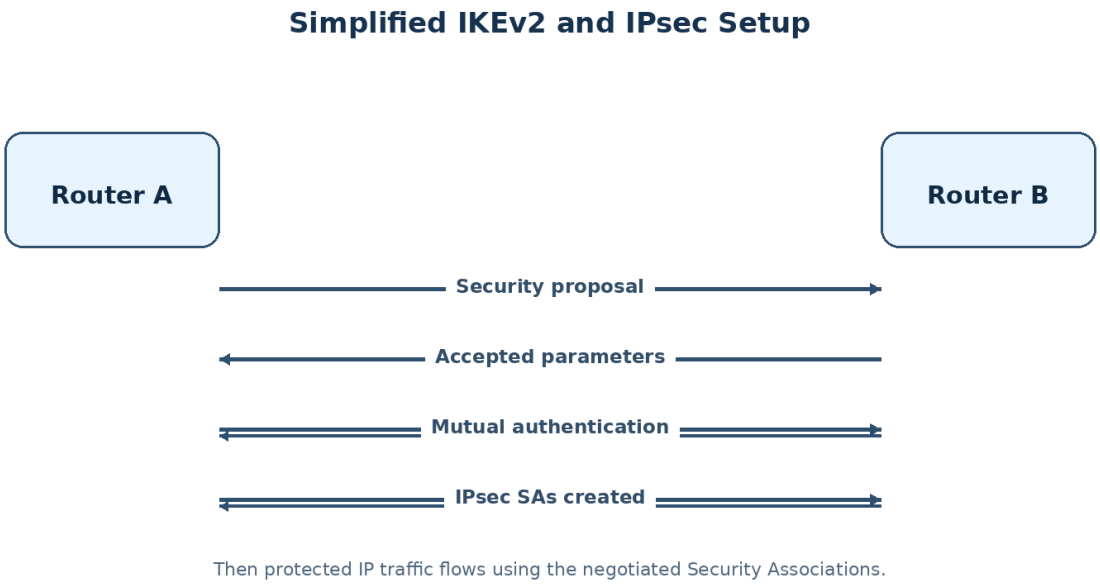
8. Security Associations and Internet Key Exchange version 2 (IKEv2)

Security Association (SA)

An SA is a one-directional logical agreement that defines how IPsec traffic is protected. Two-way communication normally requires one SA in each direction.

Direction	Example
SA 1	University Router → Branch Router
SA 2	Branch Router → University Router

What IKEv2 Does



- Authenticates the two peers.
- Negotiates security algorithms and parameters.
- Establishes cryptographic keys and creates IPsec SAs.
- Maintains the secure relationship and replaces keys when required.

Site-to-Site Example

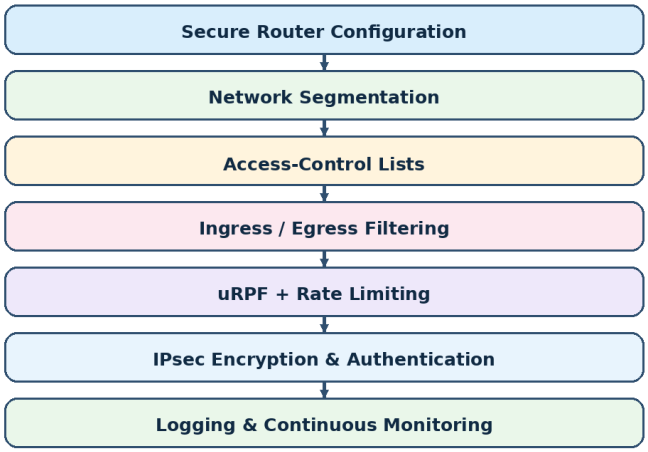
Main campus 192.168.10.0/24 ↔ encrypted IPsec tunnel ↔ branch campus 192.168.20.0/24. The gateways identify protected traffic, authenticate each other, establish SAs, encrypt packets, and decrypt them at the other site.

9. University Case Study, Summary, and Review

University Network Case Study

Requirement	Recommended protection
Students must not access router management.	Management ACL + segmentation
Internet packet claims an internal source address.	Ingress filtering + uRPF
Only required traffic reaches the grade server.	Extended ACL
Two buildings exchange private data over Internet.	IPsec tunnel mode
Suspicious Internet Control Message Protocol (ICMP) traffic appears.	Rate limiting + logging + monitoring
Routing information must be protected.	Routing authentication + router hardening

Defense in Depth at Layer 3



Key Points

- IPv4 delivers packets but does not automatically provide confidentiality or strong source authentication.
- ACLs, filtering, uRPF, segmentation, rate limiting, and router hardening reduce Layer 3 risk.
- ESP is the common IPsec choice when confidentiality is required; AH does not encrypt.
- Tunnel mode protects the entire original IP packet; IKEv2 establishes and manages SAs.

Quick Review Questions

- What is IP spoofing and why is it possible?
- What is the difference between standard and extended ACLs?
- What is the difference between AH and ESP?
- What is the difference between transport mode and tunnel mode?

- What is the purpose of IKEv2 and a Security Association?

Selected references: RFC 791; RFC 6274; RFC 4301; RFC 4302; RFC 4303; RFC 7296; NIST SP 800-77 Rev. 1; W. Stallings, Network Security Essentials.